2025 IEEE 8th International Conference on Pattern Recognition and Artificial Intelligence, Guiyang, China, August 15-17, 2025

DIGRD: Disguised Inertial Gait Recognition of Dataset and Method

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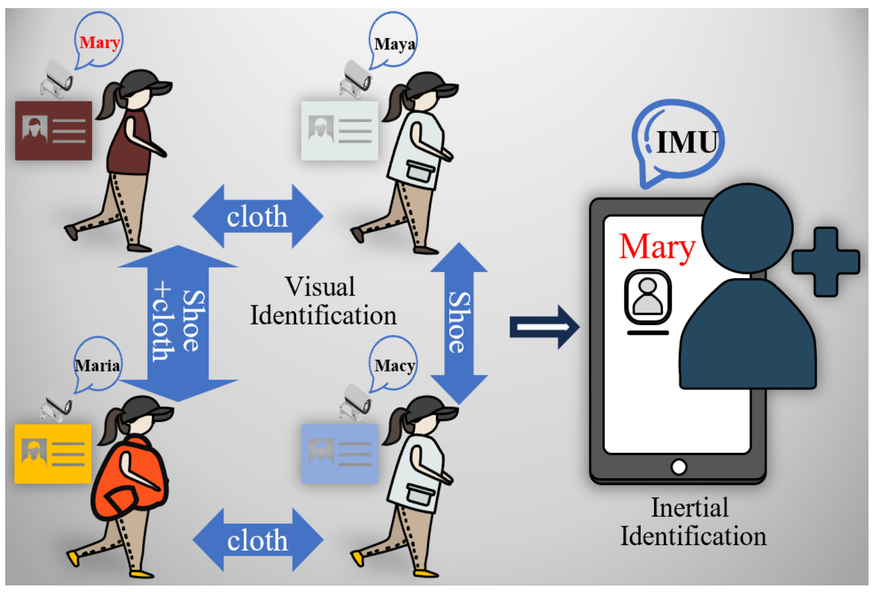
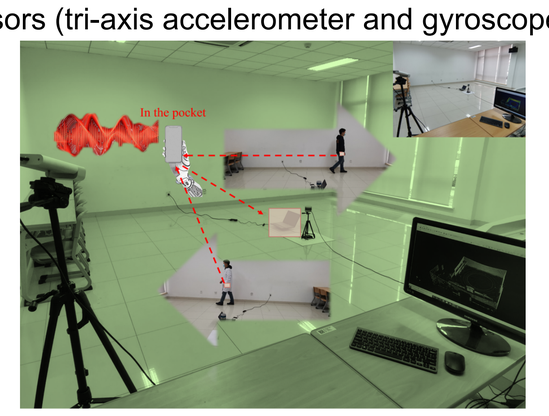
Background

- Gait recognition offers **contact-free**, **privacy-preserving** biometrics.
- Most methods use **vision-based** approaches vulnerable to **appearance changes**.
- **Clothing and footwear** disguises can alter inertial gait patterns but remain underexplored.
- We present **DIGRD**, the first benchmark for **disguised inertial gait recognition**.



Dataset

Figure 1 Participants were instructed to walk naturally around the data collection setup while carrying **a smartphone** equipped with inertial sensors (tri-axis accelerometer and gyroscope).



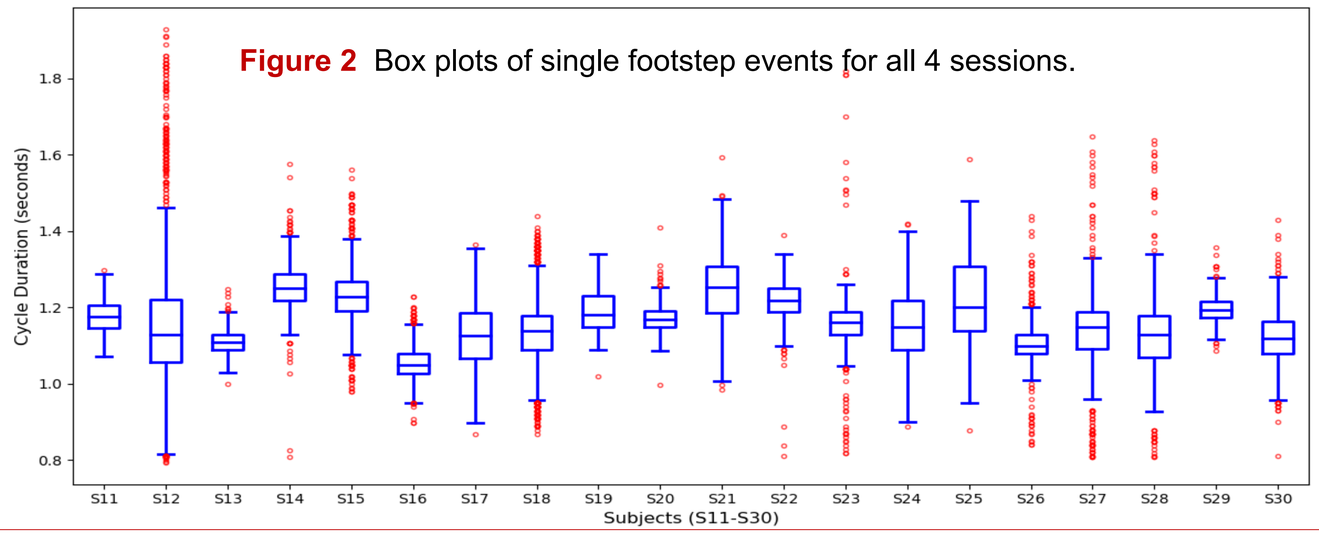
Theme I: Participants wore a thick coat and the first type of footwear, walking in a counterclockwise direction.

Theme II: Participants removed the coat (maintaining footwear type 1) and walked in a clockwise direction.

Theme III: Participants changed to footwear type 2 while maintaining the coat-less condition, walking in a counterclockwise direction.

Theme IV: Participants wore a white lab coat and walked in a counterclockwise direction.

Figure 2 Box plots of single footstep events for all 4 sessions.



Method & Results

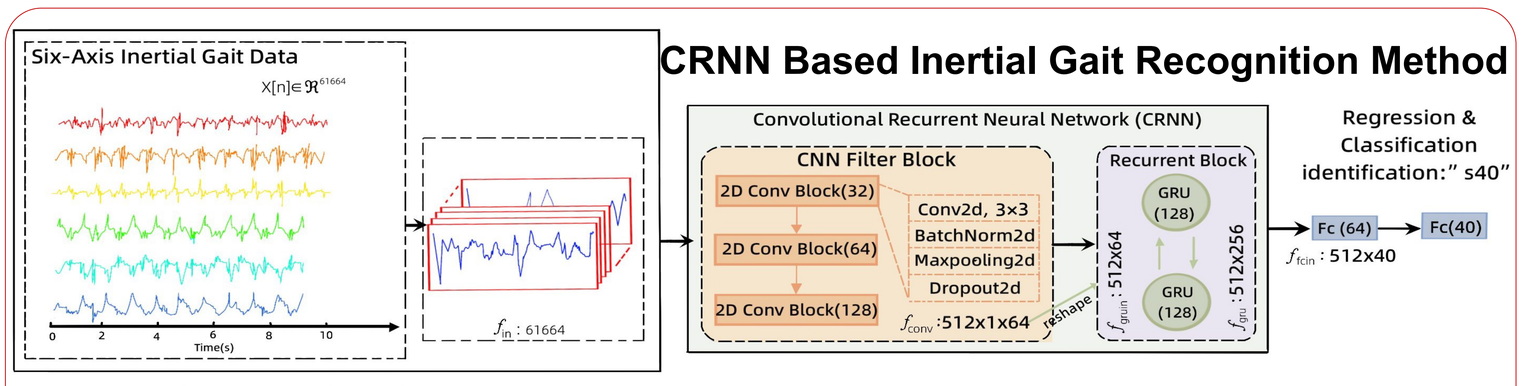


Figure 3 Overview of the **CRNN**-based inertial gait recognition method. The model extracts spatial and temporal gait features from the accelerometer and gyroscope.

Baseline Classification Results

Covariate	Setting		SVM		CNN		CN+LS		CRNN	
	Trian	Test	Acc.	Mean	Acc.	Mean	Acc.	Mean	Acc.	Mean
None	rd_1	rd_2	78.72	78.88	98.92	98.87	99.01	98.97	99.40	99.34
	rd_2	rd_1	79.04		98.77		98.92		99.27	
Clothing	I, IV	II, III	61.65	60.78	86.19	85.55	87.65	87.25	89.04	88.78
	II, III	I, IV	59.90		84.91		86.85		88.52	
Shoe Type	I, II	III, IV	56.93	55.33	83.41	81.98	85.26	83.68	86.06	84.85
	III, IV	I, II	53.72		80.55		82.10		83.64	

Ablation Experiments

Covariate	Setting		CRNN	
	Trian	Test	Acc.	Mean
None	rd_one	rd_two	99.40	99.34
	rd_two	rd_one	99.27	
Clothing	II	I	71.79	71.36
	I	II	70.93	
	IV	III	77.08	78.01
	III	IV	78.93	
Shoe Type	II	III	73.38	72.84
	III	II	72.29	
Cloth + Shoe	III	I	71.65	72.85
	I	III	74.05	
	IV	I	65.73	67.01
	I	IV	68.28	
	IV	II	78.25	79.34
	II	IV	80.43	

Other Gait Windowing Strategies

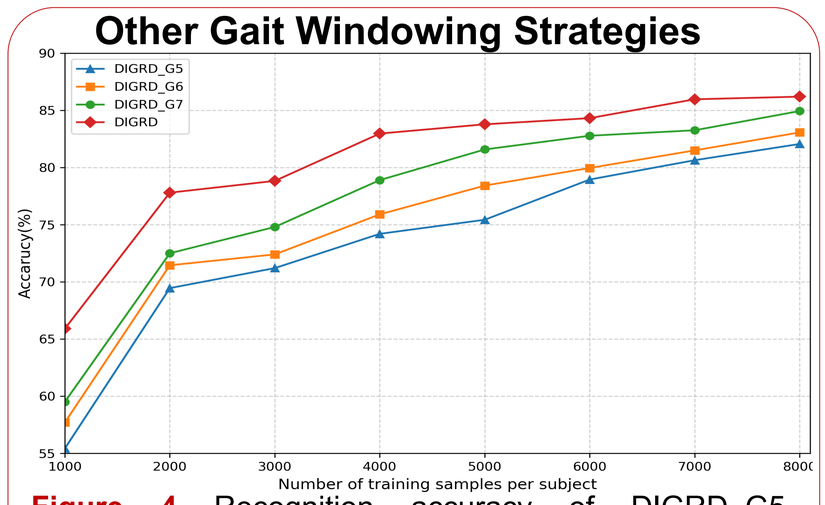


Figure 4 Recognition accuracy of DIGRD_G5, DIGRD_G6, DIGRD_G7, and the original DIGRD dataset under **various training samples.**

IMU-Based Gait Segmentation Algorithm

Algorithm 1: IMU-Based Gait Segmentation Algorithm

Input:
- $P_{\text{seg}} = \{p_i \mid i = 1, 2, \ldots, n\}$
- Thresholds for peak detection:
  - `max_threshold` = 6.0
  - `min_threshold` = 1.0
  - `peak_threshold` = 1.2
  - `threshold_roi` = 2.5

Output:
- Segmented gait data
- Start/end points of gait cycles
- Gait event points (local maxima/minima)

Step 1: Read IMU data
$p_i \leftarrow \text{readIM}(P_{\text{seg}})$;
Step 2: Noise detection
$p_{\text{raw}} \leftarrow \text{noiseDetection}(p_i)$;
Step 3: Detect local maxima
max_points $\leftarrow$ findMaxPeaks(p_{raw}["AngVel_X"], threshold = `peak_threshold`) ;
Step 4: Determine termination index
end_index $\leftarrow$ findEndPoint(p_{raw}["AngVel_X"], `max_threshold`, `min_threshold`) ;
Step 5: Filter data via ROI
Define ROI: $\{(x, y, z) \mid x^2 + y^2 + z^2 \leq \texttt{threshold_roi}\}$
$p_{\text{filtered}} \leftarrow \text{select}(p_{\text{raw}}, \text{ROI})$;
Step 6: Detect local minima
min_points $\leftarrow$ findMinPoints(p_{raw}["AngVel_X"], max_points) ;
Step 7: Extract gait cycles
for $i \leftarrow 1$ **to** *length*(*max_points*) − 1 **do**
  Extract segment [max_point[i], max_point[$i+1$]]
  if *max_point*[$i+1$] > *end_index* **then**
    Extend segment to end_index
  end
end
Step 8: Detect gait events
for $i \leftarrow 1$ **to** *length*(*max_points*) **do**
  if *i is even* **then**
    Detect events in [max_point[i], min_point[i]]
  else
    if $i > 1$ **then**
      Detect events in [min_point[$i-1$], max_point[i]]
    end
  end
end
Step 9: Validate consistency
if *length*(*max_points*) ≠ *length*(*min_points*) **then**
  Reprocess data (return to Step 2)
end
Step 10: Return results
return *Segmented data, cycle boundaries, event points* ;

Conclusions

- **DIGRD**, the first benchmark dataset designed for disguised inertial gait recognition.
- Both clothing and footwear disguises significantly reduce inertial gait recognition, with footwear reduce **to 84%**, and clothing to **87%**.

Wu S , Zhai X , Hu Z ,et al. Advanced acoustic footstep-based person identification dataset and method using multimodal feature fusion[J].Knowledge-based systems, 2023.